Nishan Kharel

Trustworthy Medical AI · Computer Vision · Large Language Models

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Research Interests

Trustworthy AI and foundation models focus on building reliable, scalable, and adaptable models for a wide range of data modalities, including text, images, speech, code, scientific data, and domain-specific datasets to develop trustworthy and adaptable AI systems.

Education

B.Sc. in Computer Science and Information Technology (B.Sc. CSIT) *Completed Feb 2026*
Texas International College, Tribhuvan University — Kathmandu, Nepal *Aggregate 80.5%*

Publications and Preprints

[P1] **Kharel, N.**, & Simkhada, P. (2026). *Does It Know When It Is Wrong? Classification versus Retrieval for Trustworthy Skin-Lesion Screening on Degraded and Real-World Images.* Preprint, Zenodo. doi:10.5281/zenodo.21188527 | ResearchGate

[P2] **Kharel, N.**, & Guragain, A. (2026). *Cancer-LLM: A PubMed Central–Derived Instruction Dataset and Fine-Tuned BioBERT Model for Multi-Class Cancer Type Classification.* Preprint, Zenodo. doi:10.5281/zenodo.20935607

[M1] **Kharel, N.**, & Simkhada, P. *A Database and Semantic Comparison System for Fire Safety Regulatory Guidance. Manuscript in preparation*, 2026–2027.

Research Experience

Trustworthy Skin-Lesion Screening under Degradation and Modality Shift *2026*
Independent research · first author · [P1]

- Compared a fine-tuned **EfficientNet-B0** classifier against a **DINOv2** embedding plus nearest-neighbour **retrieval** screener on **HAM10000** (10,015 images, lesion-grouped splits); built a degradation benchmark (blur, JPEG, low light, noise, rotation) on which the classifier's melanoma recall collapses towards zero while retrieval holds 40–50% — a failure aggregate accuracy hides.
- Validated on **PAD-UFES-20** (2,106 real smartphone photographs) with no retraining: retrieval caught **40% of melanomas vs. 11%**, and flagged the distribution shift far better (**OOD AUROC 0.94 vs. 0.72**) — the classifier failed silently.
- Distance-based isotonic-calibrated confidence reached **ECE 0.03** under degradation (classifier: 0.08 → 0.14). Reported bootstrap 95% CIs, a majority-class baseline and ablations, stating non-significant results and one negative finding honestly.

Cancer-LLM: Biomedical Instruction Dataset and BioBERT Classifier *2026*
First author · [P2]

- Built an open instruction dataset of ∼**151,800 examples** mined from **PubMed Central** across **9 cancer types** and three clinical categories (drugs, symptoms, prevention).
- Fine-tuned **BioBERT** for multi-class cancer-type classification, reaching **89.8% accuracy**; selected for the Texas International College mini-research programme and presented to external evaluators,and awarded a competitive institutional Mini Research Grant and presented to external evaluators.

Semantic Comparison System for Fire Safety Regulatory Guidance *2026 – present*
Co-author · with an M.Sc. Data Science researcher, United Kingdom · [M1]

- Co-developing clause-level alignment over **2,867 UK fire safety clauses** using **Sentence-BERT** retrieval with calibrated similarity bands, plus a rule-based check that flags changed numeric limits and modal verbs inside otherwise "similar" pairs.

Protein Function Prediction — CAFA-6 (Biomedical ML) *2025 – 2026*
Independent project

- ReFramed Gene Ontology annotation as hierarchical multi-label prediction: wrote a GO parser with ancestor propagation over **40,000+ terms** and explored **ESM-2** protein-language-model embeddings with class-weighted per-ontology classifiers.

Shishu Care — Infant Cry Classification (Healthcare Audio) *2025*
Team project

- Designed a **CNN–BiLSTM with attention** over mel spectrograms to classify infant cries (hunger, pain, burping, sleep) with near real-time inference.

Professional Experience

Applied Scientist / Research and Tech Lead *February 2026 – Present*
Samni Labs · USA (remote)

- Technical lead for **HealthCare Suite**, a live production care-plan automation system for Washington-state adult family homes: **FastAPI + LangGraph** on **AWS** (Bedrock, Transcribe Medical, S3, SQS/SNS) over **PostgreSQL/RDS**, extracting a ∼300-field clinical schema from PDFs and dictated audio with human-in-the-loop review.
- Hardened it for a healthcare buyer's security review: per-user TOTP MFA with encrypted secrets, least-privilege database roles, immutable audit trail, automatic redaction of direct identifiers before any model call, and end-to-end HTTPS.

AI Engineer (Internship) *Oct 2025 – Feb 2026*
Mantra Ideas Tech · Kathmandu, Nepal

- Engineered a computer-vision banknote recognition system for denomination classification; fine-tuned **text-to-speech** models for Nepali voice synthesis.

Data Scientist (Internship → Full-time) *Jan 2025 – Sep 2025*
Mind Risers Technology · Kathmandu, Nepal

- Built a deployed ATS / resume-matching system (Tesseract OCR, spaCy NER, Sentence-Transformer retrieval with an LLM scoring stage) and a Nepali fake-news classifier using BERT-based transformers.

Data Science Consultant (Part-time) *Feb 2025 – Feb 2026*
Namaste Network · Kathmandu, Nepal

- Taught neural networks and transformer architectures (backpropagation, attention) and designed the data-science bootcamp curriculum.

Machine Learning Tutor / Teaching Assistant *2024*
Texas International College · Kathmandu, Nepal

- Delivered **20+ lectures** on AI and machine learning and mentored undergraduates through Python and capstone project work.

Selected Technical Projects

- **Cross-SDK Propagation Agent —** fine-tuned **Qwen3-32B** with a two-stage **QLoRA** pipeline on **AWS SageMaker** (54,000 generated Q&A pairs, **75%** closed-book domain reasoning, +45 pts over base); RAG backbone with AST-based chunking (11,449 chunks), **Qdrant** and **BGE-M3** embeddings, **85%** cross-SDK bug-fix propagation in a single inference.
- **Bank Note Recognition — EfficientNet-B0** embeddings with a **ChromaDB** vector store for nearest-match retrieval; adaptive thresholding and auto-rotation for folded and degraded notes.
- **RAG Chatbot with pgvector —** production retrieval-augmented chatbot (**FastAPI + PostgreSQL/pgvector + Gemini**), containerised with Docker Compose.

Technical Skills

Languages	Python, C, SQL
ML & Deep Learning	PyTorch, TensorFlow, scikit-learn, CNNs, transformers, computer vision, NLP, model calibration, out-of-distribution detection
LLMs & Retrieval	QLoRA/LoRA/PEFT, Hugging Face, instruction-tuning, RAG, pgvector, Qdrant, ChromaDB, Sentence-BERT, BGE-M3, DINOv2, ESM-2
Systems & Cloud	FastAPI, LangGraph, PostgreSQL, Docker, Git, AWS (SageMaker, Bedrock, RDS, S3, CloudFront), GPU training
Scientific Practice	Bootstrap confidence intervals, ablation studies, imbalanced and multi-label evaluation, LaTeX

Grants, Awards and Certifications

- **Mini Research Grant**, Research Learning and Development Department, Texas International College, 2026 — one of six proposals funded from an open institutional call following internal and external review. Awarded project: *Cancer-LLM* [P2].
- **2nd place**, *Code for Change* hackathon; participant in Kings College

References

Saroj Ghimire	**Manish Singh**	**Pukar Upreti**
Senior Lecturer	Senior Cloud Engineer	Software Engineer
Texas Int'l College	Amazon	Bunnings, Australia
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